Geometry and Stability in the Compact Spherical Tokamak

Negative Triangularity and High-Beta Confinement

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Abstract

The performance and stability of tokamak plasmas are strongly influenced by their geometric shaping. In compact spherical tokamaks (low aspect ratio), advanced shaping configurations featuring high elongation and negative triangularity (an inward-facing "D" shape) have demonstrated improved confinement and stability at high plasma pressure (high beta). This paper analyzes the role of geometry in achieving stable, high-beta operation, with emphasis on negative triangularity effects observed in experiments on devices like TCV and DIII-D. We formulate the equilibrium framework via the Grad–Shafranov equation and define key shaping parameters—elongation (κ), aspect ratio (A), triangularity (δ)—along with the normalized beta (β_N) used to gauge pressure limits. The impacts of extreme shaping on vertical stability and bootstrap current fraction are examined. Negative triangularity plasmas ($\delta < 0$) are shown to suppress edge instabilities (notably avoiding ELMs) while maintaining H-mode-level confinement without an edge pedestal[1][2]. High-beta regimes in spherical tokamaks ($A \approx 1.5$–2.0, $\kappa > 2.5$) are explored, including the Troyon limit on β and the engineering constraints of REBCO high-field magnets. Structural considerations such as Lorentz force management via bucking cylinders and strain-focused magnet design (as in the Kronos S.M.A.R.T. tape architecture) are discussed in enabling such shapes. Finally, we consider the relevance of these geometries to aneutronic fusion (D–He^3, p–^{11}B), where shaped field-lines can facilitate direct energy conversion of charged fusion products. Negative triangularity and compact geometries emerge as promising strategies for next-generation fusion reactors, combining high confinement efficiency with manageable plasma–wall interactions and advanced fuel compatibility.

Table of Contents

Introduction 4
Theoretical Framework and Equilibrium Analysis 5
Negative Triangularity Configuration 8
High-Beta Operation and Limitations 11
Structural and Magnetic Design Considerations 14
Relevance to Aneutronic Fuel Cycles 16
Risk Acknowledgment and Ongoing Research Disclosure 18
Conclusion 19
BIBLIOGRAPHY 21
CHARTS 24
FIGURES 25

Introduction

Geometric shaping of the plasma cross-section is a critical lever for improving tokamak stability and confinement. Decades of research have shown that elongated, non-circular plasmas can achieve higher pressure and better energy confinement than circular ones, albeit with added control challenges. Triangularity, the "D-shape" indent of the plasma, also has a profound effect: when the flat side of the D faces the tokamak center (positive δ), plasmas tend to have a high edge pressure pedestal and strong edge instabilities; when the curved side faces the center (negative δ), the plasma cross-section is inverted, yielding surprisingly different behavior[3]. Recent experiments indicate that negative triangularity can *suppress* deleterious instabilities that expel particles and energy, thereby preventing transient wall loads[4]. In fact, plasmas with $\delta < 0$ exhibit a lower level of turbulence and are less prone to edge-localized modes, yet they maintain energy confinement comparable to standard high-confinement modes[1]. Garcia-Muñoz *et al.* note that negative triangularity "has a lower level of fluctuations inside the plasma, but it also has a larger divertor area to distribute the heat exhaust," making it attractive for compact reactor designs[5]. These benefits motivate exploring unconventional shapes in tandem with the **spherical tokamak** approach.

Spherical tokamaks (STs) operate at a low aspect ratio (A = R₀/a ≲ 2), yielding a squat, cored-apple plasma shape instead of the conventional doughnut. This more "spherical" plasma provides strong self-stabilizing curvature and high beta capability. The ST geometry "makes it easier to confine the plasma," as evidenced by devices like NSTX and MAST achieving higher pressure fractions than conventional tokamaks[6]. Indeed, NSTX attained a toroidal beta approaching 40% – roughly four times the beta of similarly equipped conventional tokamaks – by leveraging strong shaping and feedback control[7]. High beta (β) is desirable for fusion power density and is especially critical for advanced fuels. For example, the D–^{3}He reaction demands higher plasma temperatures and pressure; accordingly, reactor studies have identified that "high beta and/or high field innovative confinement concepts" (like spherical tokamaks or field-reversed configurations) are needed to make D–^{3}He burning feasible[8]. The prospect of aneutronic fusion also brings an additional incentive: most of the fusion energy emerges in charged particles (α or p), which can be converted directly to electricity[9]. A compact, high-β fusion core with minimal neutron yield could thus integrate novel *direct energy conversion* systems, avoiding the efficiency penalties of a purely thermal power cycle. These opportunities drive the interest in combining **compact geometry, high elongation, and negative triangularity** shaping to maximize stability, confinement, and reactor practicality.

This paper provides a focused review of how such shaping strategies impact plasma equilibrium and stability, drawing on both theoretical models and experimental results. We first outline the axisymmetric equilibrium framework (Grad–Shafranov equation) and define the shaping parameters and normalized beta used to quantify performance limits. We then examine negative triangularity configurations in detail, summarizing the experimental evidence of improved confinement and edge stability (e.g. the ELM-free regimes on TCV and DIII-D). Next, we discuss high-beta operation in spherical tokamaks, including the distinction between poloidal, toroidal, and normalized beta, and the empirical Troyon limit that often constrains β. We address how far low-aspect-ratio devices have pushed β_N and what limits arise (e.g. resistive wall modes, coil stresses). Engineering aspects are considered in Section 6: the intense Lorentz forces in low-A magnet systems demand innovative structural solutions (bucking cylinders, wedged supports) and advanced superconductors. We highlight a "strain-first" coil design philosophy, exemplified by the Kronos tape architecture, which prioritizes maintaining HTS tapes below their strain damage threshold. Finally, Section 7 discusses why these geometric choices are well-suited for aneutronic fuel cycles—how shaping can aid in channeling charged fusion products and managing heat and particle exhaust in the absence of a neutron-dominated blanket. The goal is a holistic understanding of how geometry and stability considerations intertwine in next-generation compact fusion tokamaks.

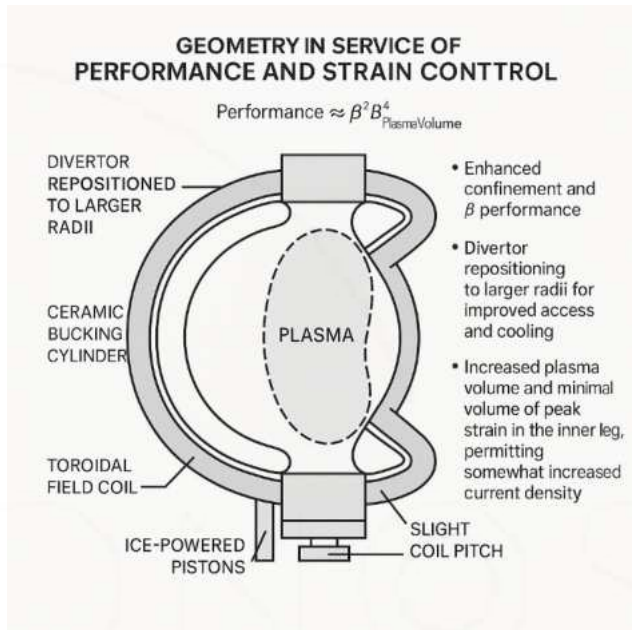


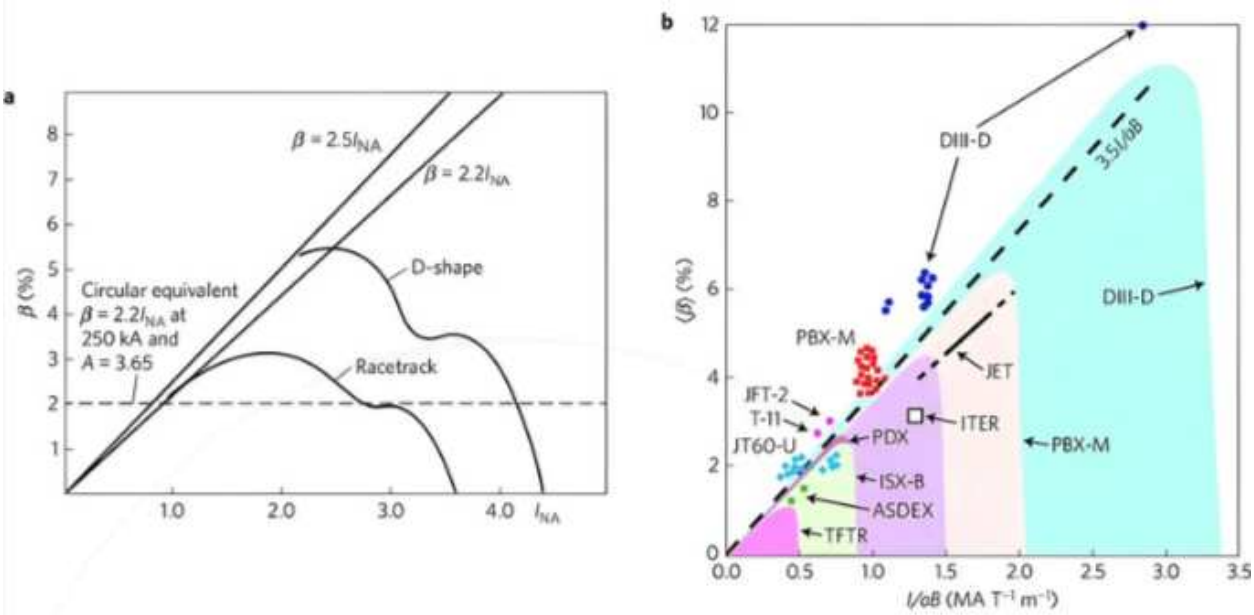
Figure 1: Engineering solutions in compact low-A magnets: bucking cylinder, coil wedge supports, and pre-tensioning elements that accommodate high Lorentz forces and minimize strain at 30–35 T operation.

Theoretical Framework and Equilibrium Analysis

Plasma equilibrium in an axisymmetric tokamak is governed by the Grad–Shafranov equation, a two-dimensional elliptic partial differential equation that describes how the poloidal magnetic flux (ψ) is distributed inside the torus. In essence, it balances the inward Lorentz force on the plasma current against the outward pressure gradient and tension of magnetic field lines. The Grad–Shafranov formulation makes explicit how shaping coils and plasma current profile determine the flux surfaces' shape. The equilibrium condition can be expressed (in cylindrical coordinates) as $\Delta\psi = -\mu_0 R^2\, dP/d\psi - (1/2) dF^2/d\psi$*, where P(ψ) is the plasma pressure profile and F(ψ) relates to toroidal current. Although we omit the detailed equation derivation here, its key takeaway is that plasma cross-sectional shape (elongation, triangularity, etc.) enters through boundary conditions and source terms in this equation. Solutions to Grad–Shafranov yield nested flux surfaces that can be highly non-circular.

Shaping parameters provide a quantitative handle on the plasma geometry. The **aspect ratio** $A = R_0 / a$ is the ratio of the major radius to minor radius of the plasma. Conventional tokamaks have A ~ 3–4, whereas spherical tokamaks push A down to ~1.5–2.0 (a very "fat" torus). **Elongation** κ describes the vertical stretching of the plasma cross-section; κ is the ratio of the plasma's height to its width (or of the vertical minor radius b to horizontal a). Higher elongation allows a greater plasma current and pressure for the same toroidal field, but also leads to a thinner, vertically extended plasma column that is more susceptible to vertical motion. **Triangularity** δ characterizes the shaping of the plasma's triangular indent. By convention, δ is positive when the plasma's upper/lower shape indents toward the inboard side (giving the cross-section an outboard-pointing "D" shape). If instead the plasma cross-section bulges outward on the inboard side and is indented on the outboard side, δ is negative[3]. In practice, δ is measured as the horizontal displacement of the plasma's top or bottom relative to the centerline, normalized by the minor radius. Moderate positive triangularity (δ ~ 0.3–0.5) is standard for achieving high pedestal pressure in H-mode, whereas δ < 0 creates an "inverted D" shape that curtails the pedestal but can improve core stability (as discussed in Section 4).

A crucial performance parameter is the plasma **beta (β)**, the ratio of plasma pressure to magnetic pressure. **Toroidal beta** (often just "β") is defined as the volume-averaged plasma pressure divided by the pressure of the toroidal magnetic field (B_T) on axis[10]. It essentially measures how efficiently the magnetic field is utilized to confine plasma—higher β means more fusion output for a given magnetic field strength. **Poloidal beta** (β_p) is similarly the ratio of plasma pressure to the poloidal field pressure; it relates to how the plasma current and pressure are partitioned (and is proportional to the no-wall ideal MHD stability criterion in some cases). Because tokamaks can vary widely in size, field, and current, it is useful to define a **normalized beta** β_N, which is dimensionless. One common definition is β_N = β (%) a [m] B [T] / I_p [MA]*, which roughly scales out the dependence on machine parameters. The empirical Troyon limit** states that β_N must remain below an empirically observed constant (around 3–4 for conventional aspect ratios) for stable operation without wall stabilization. In other words, there is a maximum pressure a tokamak can stably confine for given B, size, and current; exceeding that tends to trigger macroscopic MHD instabilities (like ballooning modes or kink modes) unless advanced control or conducting walls are used.



Charts 1: Plasma shaping and beta performance. (a) Comparison of normalized beta (β) versus normalized current (I_n/NA) for different plasma cross-sections—circular, D-shape, and racetrack—showing the higher stability limits achievable with optimized shaping. (b) Experimental database of beta versus I_paB across major tokamaks (JET, ITER, DIII-D, PBX-M, ASDEX, etc.), illustrating the path toward advanced high-β operation and confinement scaling.

Shaping strongly influences these stability limits. Higher elongation κ allows achieving a higher poloidal beta (and thus total beta) because the plasma cross-section can carry more current and pressure before reaching stability boundaries. In fact, energy confinement typically improves with elongation – an empirical scaling suggests τ_E ∝ κ^^{0.7}, meaning a 50% taller plasma might gain ~30% in energy confinement time[11]. However, elongation comes at the cost of **vertical stability**. A plasma with κ > 1 is prone to develop an axisymmetric n=0 instability: it can "tilt" or move vertically out of the magnetic well. In an elongated cross-section, the upward and downward restoring forces balance precariously, and without a conducting wall nearby or active feedback, the plasma can rapidly drift into the wall ("vertical displacement event"). The maximum elongation is often limited by the capability of feedback control systems and the presence of a close-fitting wall. Analyses of the axisymmetric stability (resistive wall mode, RWM) show that increasing κ eventually triggers a growth rate that cannot be controlled by a given system, defining κ_{max} for that device[12]. For instance, in DIII-D experiments, negative triangularity plasmas were run at κ ≃ 1.3 (lower than the usual κ ≃ 1.7 for positive δ H-modes) because higher elongation would have been unstable without additional measures[13]. Thus, while high κ is desirable for confinement, it must be balanced against vertical stability limits.

Triangularity δ also affects stability and current distribution. Positive δ plasmas tend to have better coupled plasma–wall stability for the vertical mode (because the plasma is "leaning" on the inboard side where a wall or stabilizing plate can be placed). Negative δ, on the other hand, can destabilize the n=0 mode unless the wall shape and plasma positioning are carefully considered[14][15]. Interestingly, one study using the AVSTAB code found that a *non-conformal* wall (e.g. a wall shaped for a positive δ plasma enclosing a negative δ plasma) could provide some stabilization, as long as the plasma wasn't too close to the wall at the triangularity nulls[16]. This

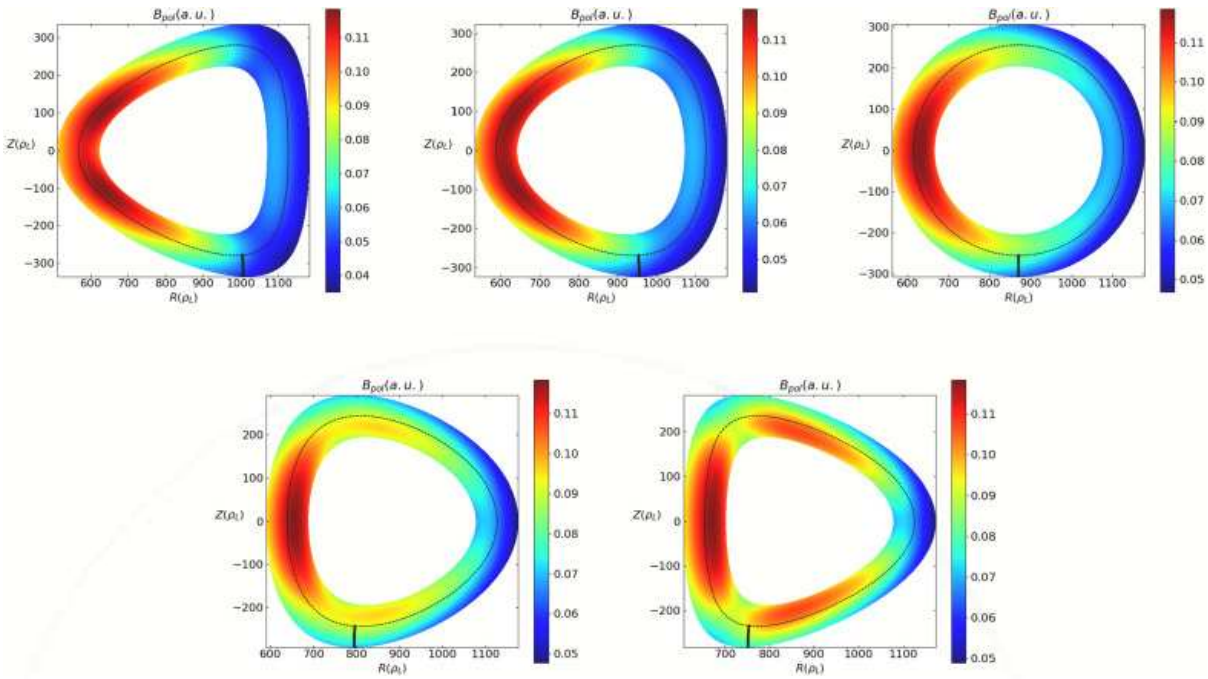
indicates that engineering the vacuum vessel shape and feedback coil locations is part of the stability solution for negative triangularity configurations.

Another key impact of shaping is on the **bootstrap current**, a self-generated plasma current driven by pressure gradients. In a torus, collisions in a pressure gradient cause a preferential flow of particles along the field lines, effectively "bootstrapping" a current. The fraction of plasma current provided by this effect increases with beta and with broad pressure profiles. Shaping plays a role by modifying the geometry of flux surfaces and the distribution of pressure. Spherical tokamaks, in particular, capitalize on very high β and strong shaping to maximize bootstrap current fraction[17]. Because of its shape and high β, NSTX, for example, could drive up to ~60% of its plasma current by a combination of bootstrap and neutral beam injection (leaving only ~40% to be driven inductively)[18]. This greatly eases the burden on solenoid flux for steady-state operation. Negative triangularity tends to have a broader, more gradational pressure profile (with no sharp pedestal), which can actually enhance core bootstrap current while reducing edge bootstrap current (since the steep edge gradient is absent). Theoretical work has suggested that negative-δ flux surfaces lead to an enhanced total bootstrap current for a given pressure profile[19]. However, a large bootstrap current can also drive certain MHD modes (like peeling-ballooning or neoclassical tearing modes), so the coupling between shaping, bootstrap, and stability is subtle. In advanced tokamak regimes, careful alignment of pressure and safety factor profiles (q profiles) is done to avoid deleterious MHD, taking into account how shape affects the bootstrap current distribution across the plasma radius.

In summary, the theoretical framework shows that compact, strongly shaped tokamaks operate in a narrower stability margin than conventional shapes, but offer higher performance. By pushing aspect ratio low and elongation high, spherical tokamaks achieve high β and strong self-driven currents, at the expense of requiring active stabilization of vertical modes. Negative triangularity introduces a new dimension to shaping control, altering edge stability and transport properties in ways that standard theory had not predicted (Section 4). We now turn to the empirical evidence and benefits of $\delta < 0$ configurations.

Negative Triangularity Configuration

The negative triangularity (NT) tokamak configuration has garnered significant interest after experiments in the late 1990s and recent years revealed remarkable performance benefits[20]. In an NT plasma, the cross-section looks like a "reverse D", with the plasma leaning away from the device center at the midplane. Initial studies on the TCV tokamak (Switzerland) found that NT L-mode plasmas achieved **improved energy confinement** relative to equivalent positive triangularity discharges, and this improvement persisted as triangularity became more negative (experiments explored down to $\delta \approx -0.5$)[21]. The confinement enhancement was linked to reduced turbulent transport: core fluctuation levels were measured to be significantly lower in NT shape, leading to steeper core pressure gradients without provoking instability[20]. Essentially, NT plasmas attained high core pressure in an L-mode-like state (no transport barrier at the edge), which was a surprising and encouraging result.



Charts 2: Magnetic flux surface contours for compact, high-elongation plasmas with negative triangularity. The color maps illustrate normalized poloidal field strength (Bθ) across various shaping configurations, showing how negative triangularity modifies edge stability, reduces ELM activity, and supports enhanced confinement in the S.M.A.R.T. geometry.

Follow-up experiments on DIII-D validated these findings in a larger, higher-performance tokamak. Marinoni *et al.* and Austin *et al.* reported **H-mode equivalent confinement in NT plasmas with no ELMs and no edge pedestal**[1]. In DIII-D, δ was varied from +0.4 (standard D-shape) down to –0.3/–0.4 in otherwise similar conditions. At δ ≈ –0.3, the plasma remained L-mode (no sudden pedestal formation) yet achieved the same thermal energy as a standard ELMing H-mode at positive δ[22]. The NT discharge had **normalized beta** up to β_N ~ 2.8 and sustained it without edge instabilities[1]. Crucially, all edge-localized modes (ELMs) vanished in the negative δ shape – even at high heating power and plasma current, no ELMs were observed[23]. This robust ELM avoidance has now been demonstrated over a wide range of conditions. A 2023 study on diverted NT discharges in DIII-D showed that strong negative triangularity "robustly suppresses all ELM activity" while still allowing substantial edge pressure gradients to form (though not as steep as a true H-mode pedestal)[2]. In these experiments, the entire edge pressure profile stayed at or below the ideal infinite-n ballooning stability limit, consistent with the theoretical prediction that NT's altered magnetic shear prevents access to the usual ELM-triggering regime[2]. In effect, NT shaping inherently capped the edge pressure gradient before an instability could erupt – an elegant way to avoid ELMs without needing external mitigation like magnetic perturbations.

The absence of ELMs in NT plasmas yields obvious benefits for **plasma–wall interactions**. Large ELMs in conventional H-modes can expel a significant fraction of plasma stored energy in milliseconds, causing impulsive heat loads on divertor plates. By operating in an ELM-free regime, an NT tokamak would spare the wall from these cyclic bursts, potentially improving divertor component lifetimes. Moreover, as mentioned, NT shapes tend to distribute heat exhaust more broadly. When δ is negative, the magnetic X-point on the divertor moves outward radially, effectively lengthening the divertor leg and increasing the area over which the scrape-off layer (SOL) plasma contacts the target plates. Garcia-Muñoz noted that negative triangularity provides "a larger divertor area to distribute the heat exhaust" compared to standard shapes[24]. In practice, experiments have seen that the heat flux footprint in NT configurations is expanded and peak heat flux reduced for comparable power flows[25]. This is a critical advantage for handling the high power densities in a reactor: the more one can spread out and mitigate the exhaust power, the easier it is to avoid damaging materials.

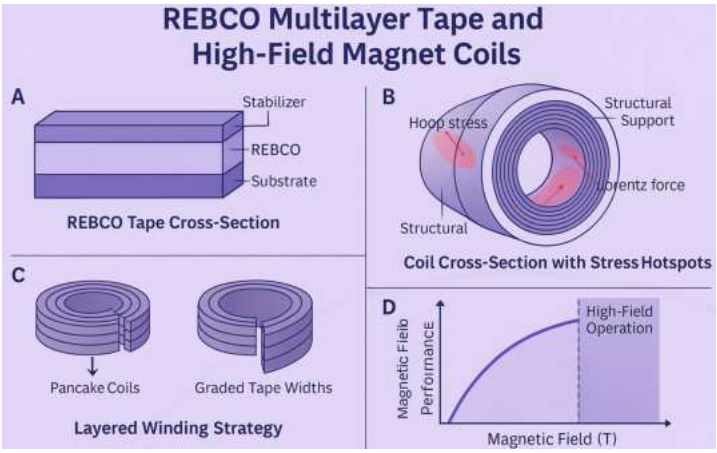


Figure 2: Negative triangularity plasma configuration in a compact spherical tokamak. The inverted-D cross-section improves edge stability, suppresses ELMs, and maintains high-β confinement suitable for aneutronic fuel operation.

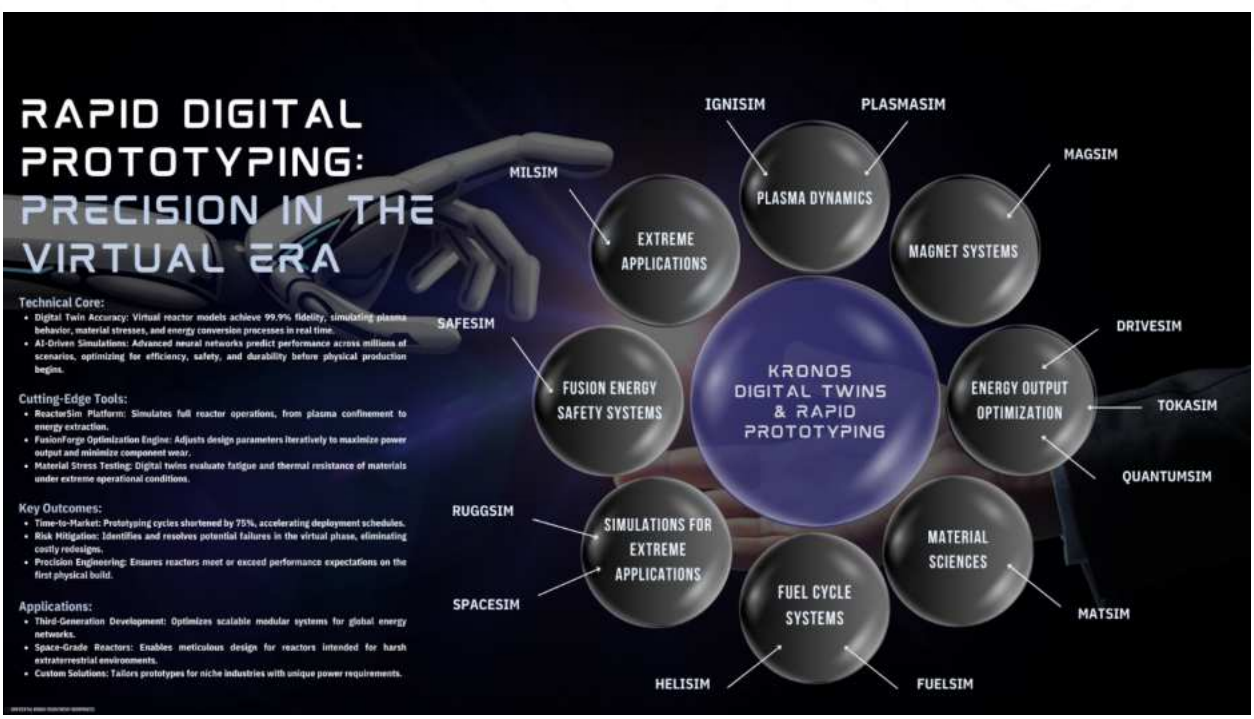


Figure 3: Divertor heat-flux footprint for negative triangularity versus positive triangularity plasmas. The strike-point profile in δ < 0 cases is broader and the peak heat load reduced, improving divertor survivability at reactor-relevant power exhaust conditions.

Another intriguing aspect of NT plasmas is the **edge turbulence behavior**. Even though NT prevents the formation of a high pedestal, it does not suffer from poor confinement. Instead, improved core confinement compensates. Detailed measurements on TCV and DIII-D found that turbulence (measured by density and temperature fluctuation levels) was 10–50% lower across much of the minor radius in NT discharges versus shape-matched positive δ discharges[26]. The leading hypothesis is that the reversed triangular shape changes the magnetic shear and curvature in the edge region in a way that is unfavorable for drift-wave turbulence, specifically stabilizing ion temperature gradient (ITG) modes or trapped electron modes that typically drive transport[27]. Additionally, NT seems to access a second-stability regime for ballooning modes differently. In conventional high-β plasmas, reaching the second stability (where pressure-driven modes are stable) often requires a certain edge current or pressure profile. NT shapes tend to *close off* access to the second stability at the edge altogether[28][29]. This might sound detrimental, but paradoxically it is beneficial: by disallowing the large pedestal (which relies on second stability), NT ensures the edge stays in first stability with a benign pressure gradient – hence no ELMs[30]. The core, meanwhile, can still achieve high pressure because turbulence is low. The result is a kind of "H-mode without a pedestal," as some have described it.

The **fusion performance** achieved in negative triangularity regimes has reached reactor-relevant levels in some parameters. DIII-D's NT discharges attained β ~ 2.8% (absolute) with β_N near 3, and confinement H-factors ~1 (i.e., H-mode grade energy confinement)[31]. Importantly, these were inner-wall-limited or slight divertor plasmas without sophisticated wall conditioning, indicating the potential for even better performance if optimized. On TCV, NT L-modes have surpassed the confinement of even H-mode reference plasmas at similar heating power[32]. There is ongoing work to explore NT in spherical tokamaks as well (MAST Upgrade plans to operate $\delta < 0$, and the new SMART tokamak in Spain is dedicated to NT studies[33]). Initial results from MAST-U's first NT experiments showed the same ELM elimination, although some differences in turbulence behavior were noted compared to conventional aspect ratio cases[34][35]. These will be important to resolve as they indicate potential nuances when combining NT with low aspect ratio (we address this in Section 5).

In summary, negative triangularity has proven to: (1) **Mitigate edge instabilities** – NT shapes naturally suppress ELMs across a broad operating window[2], offering a steady exhaust scenario; (2) **Improve core confinement** – by reducing turbulent transport, NT yields higher pressure for given input power[22]; (3) **Alter power exhaust geometry** – the divertor topology in NT spreads heat flux and may ease handling of plasma-material interaction[24]; and (4) **Remain compatible with high beta** – experiments achieved high normalized beta without encountering new limiting instabilities unique to NT, aside from the standard vertical/RWM considerations. These features make $\delta < 0$ plasmas a compelling candidate for next-generation reactors, potentially allowing operation in a regime that is high-performance like H-mode but quiescent like L-mode. The next section discusses how these high-beta regimes are attained and what fundamental limits must be respected.

High-Beta Operation and Limitations

Achieving and sustaining **high beta** is a central goal for an economic fusion reactor, since fusion power scales with the pressure squared while magnet costs scale roughly with field squared.

Spherical tokamaks have been especially successful in reaching high β due to their favorable geometry. It is useful to distinguish the various beta measures in this context:

- **Toroidal beta (β_T)** – The ratio of the volume-averaged plasma pressure ⟨p⟩ to the magnetic pressure of the toroidal field B_T on axis: β_T = $2\mu_0$⟨p⟩ / B_T^2 (often expressed as a percentage). NSTX famously demonstrated β_T ≈ 40%[7], meaning the plasma pressure was 0.4 times the confining B-field pressure – an extremely high value far beyond conventional tokamak operation (which typically maxes out around 10% in devices like DIII-D or JET). Such high β_T in NSTX was enabled by strong neutral beam heating (~7 MW) and careful plasma shaping to avoid disruptive instabilities[36].
- Poloidal beta (β_p) – The ratio of plasma pressure to poloidal magnetic field pressure. β<sub>p is less directly measurable but is related to the plasma internal inductance and current profile. High β_p means the plasma current is largely sustained by pressure-driven currents (bootstrap) rather than by external magnet flux change. Advanced tokamak regimes aim for β<sub>p> 1, indicating significant bootstrap current fraction. Spherical tokamaks often operate in high β<sub>p regimes; for instance, an ST scenario might have β<sub>T = 40% and q_{95} ≈ 12–15, implying β<sub>p on the order of 1–2 (since q ∝ (ε β<sub>p)$^{-1}$/2 roughly).
- **Normalized beta (β_N)** – As defined earlier, β<sub>N **= β (%) * a [m] * B [T] / I<sub>p** [MA]. This scale-invariant measure is useful to compare discharges across machines. The Troyon limit is often cited in terms of β<sub>N**: for conventional aspect ratio tokamaks without a conducting wall, β<sub>N** ≈ 3.5–4 is an empirical stability boundary. Spherical tokamaks, by virtue of different shaping and profile optimization, have achieved higher β<sub>N**. Experiments on the START ST achieved β<sub>N** ~ 5–6, and NSTX reached β<sub>N ≈ **7–8.5 in high-β plasmas (e.g. a discharge with β<sub>T** = 40%, I<sub>p ~ **0.8 MA, a = 0.65 m, B<sub>T** = 0.45 T gives β<sub>N** ~ 8)[37]. These values exceed the Troyon limit for standard aspect ratio, hinting that either the ST stability limit is higher or active control/fast plasma rotation provided additional stabilization.

The Troyon scaling itself must be revisited for low-A tokamaks. The original limit (β (%) ≤ 2.8 * a [m] * B [T] / I<sub>p [MA]) was derived from multi-machine data mostly at A ~ 3. For spherical tokamaks, the ideal MHD limit can be higher due to favorable shaping (high κ, high shear). However, new limiting factors appear: pressure-driven kink/ballooning modes and resistive wall modes (RWMs). High β ST plasmas are often at high q (to avoid low-q sawteeth or kinks) and rely on a conducting wall plus feedback coils to stabilize the RWM. NSTX, for example, used a set of external coils to actively counteract the growth of n=1 kink/RWM instabilities at β<sub>N above ~5[38]. With these measures, NSTX could sustain β<sub>T ~ 35–40% for many energy confinement times, limited not by gross instability but often by gradual β decreases or minor events. In the advanced ST regimes, one sees a trade-off between q-profile, bootstrap, and β**: operating at very high β (and thus high bootstrap fraction) can lead to hollow current profiles that trigger neoclassical tearing modes (NTMs), which then cap the achievable beta. Therefore, maintaining high β requires careful current profile tailoring, sometimes by reducing the bootstrap peaking (e.g. via broader pressure profiles or current drive).

Compact high-β plasmas also face engineering limits. In devices using high-field REBCO (HTS) magnets, the magnetic field and current density are pushed to extremes to confine high-pressure plasma in a small volume. There is a practical upper bound where the magnet structure can no longer tolerate the forces from a higher plasma pressure. Even if MHD stability would allow higher β, the stress on the toroidal field coils and vacuum vessel during a disruption at that β could be unacceptable. Thus, designers include a margin: for example, a private fusion company designing a high-field spherical tokamak might limit β<sub>N to ~5 even if theoretically 8 is possible, to reduce disruption risk to the magnets. Additionally, there is the thermal load – high β plasma means high fusion power density, which could overheat first-wall components if not sufficiently cooled or if divertor design is not adequate. REBCO tape magnets bring another consideration: strain limits**. The REBCO superconductor can tolerate on the order of 0.3–0.5% tensile strain before it irreversibly loses superconducting capability[39]. A very high-beta plasma exerts large electromagnetic forces on the coils (e.g., during transients or if plasma control is lost), which could deform the magnet beyond safe strain. Thus, in "REBCO-limited" systems, one effectively has a ceiling on β to ensure coil stresses remain in the elastic regime. For instance, if an ST reactor is built with HTS coils close to the plasma, it may operate at a slightly reduced β to ensure that even worst-case disruptions do not push coil strain above ~0.4%. This is not a hard physics limit but a prudent engineering cap.

Empirically, spherical tokamaks have demonstrated the feasibility of high β operation up to the limits imposed by available heating power and magnet capability. The NSTX achieved its target β ~40% at B ~0.35–0.5 T by using 6–7 MW of NBI heating[36] and strong shaping (κ ~ 2.2, δ ~ 0.4). By "carefully controlling the shape of the plasma to avoid violent instabilities," NSTX was able to sustain these pressures without disruption[40]. Later, NSTX-U aimed to extend pulse lengths and β sustainment, exploring regimes of β<sub>N > 6 and high bootstrap fraction f<sub>BS ~ 70–80%. Those experiments highlight that at extremely high β, one often encounters soft limits like intermittent small ELMs or confinement degradation due to microturbulence. There is evidence that beyond a certain β, energy confinement in STs deviates from standard scaling (possibly due to electromagnetic turbulence effects at high β). Thus, the path to higher β is not only blocked by sudden instability, but also by diminishing returns in confinement.

Finally, in the context of negative triangularity and high β, initial studies (both theoretical and experimental) are examining whether NT plasmas can reach similar β limits. One concern raised was that NT, by closing the second-stability access for ballooning modes, might limit the maximum β (since the plasma can't leverage second stability to go higher)[41]. If true, NT reactors might trade off some attainable β for the benefit of no ELMs. However, DIII-D and TCV results so far indicate NT can reach at least β<sub>N** ~ 3 without wall stabilization[1], and theory suggests that with a conducting wall, NT could go higher before hitting an ideal limit. MAST-U's NT experiments encountered some long-wavelength modes at high β (kinetic ballooning modes)[42], but it's an open research area. Regardless, any advanced tokamak, NT or not, will likely be designed with β well below the true ideal limit for a safety margin.

In summary, high-beta operation in compact tokamaks is a balancing act between maximizing pressure for performance and staying within stability/engineering limits. Spherical tokamaks have proven high β is achievable (with β<sub>N** up to ~8 in pulses)[37], and negative triangularity offers a new route to maintain stability at high β without ELMs. The next considerations are

whether the machine structure can sustain the forces of these plasmas and how to design magnets and supports accordingly, which we address next.

Structural and Magnetic Design Considerations

Extreme shaping and high-field operation in a compact tokamak put stringent demands on magnet and structural design. The toroidal field (TF) coils in a low-aspect-ratio device have a tight curvature around the small central column, which leads to very large **Lorentz forces**. Each TF coil experiences a strong outward (hoop) force trying to expand it away from the torus center, as well as a net compression against the central structure. In high-field designs, the magnetic pressure ($B^2/2\mu_0$) can be hundreds of MPa, so the coil structure must withstand forces in the tens of meganewtons. To manage these forces, designers often use a central **bucking cylinder** or tie-rod structure. In this scheme, the inward force of the TF coil inner legs is reacted by a robust central cylinder (which can be the central solenoid or a separate structure) that "bucks" the coils, preventing them from collapsing inward. By having the TF inner leg press against a bucking cylinder, one can significantly reduce the peak stress in the coil material[43]. Essentially, the cylinder provides an equal and opposite force, cancelling some of the tension that would otherwise strain the coil. The ARIES reactor studies in the 1990s, for example, implemented a bucking cylinder as a key element to keep TF coil stress within allowable limits at high field[43]. Spherical tokamaks take this further: the central column (which houses TF coil bundle and possibly the solenoid) often serves as a massive support that all TF coils latch onto.

Another structural strategy is careful control of **coil geometry and supports** – sometimes termed coil "pitch" and wedging. TF coils are usually D-shaped; if they are perfectly up-down symmetric, the net vertical force on each coil cancels. However, any asymmetry (such as the presence of a significant triangularity or if the coil is not exactly planar) can introduce vertical forces that try to push the coil legs apart. Engineers include features like **shear keys, intercoil structures, and flange bolts** at the top and bottom of coils to hold them together against these forces. In spherical tokamaks, space is limited, so these structures often double as vacuum vessel ribs or neutron shielding. For example, NSTX had a massive outer support structure tying the TF coil upper sections together to resist out-of-plane bending. When triangularity is varied (e.g., going to $\delta < 0$), the shaping field coils (PF coils) that pull the plasma outward put additional load on the TF coil cases and the bucking cylinder. Thus, the entire magnet system must be designed to accommodate a range of shapes without yielding. A **"strain-first" design philosophy** has emerged for modern superconducting magnets: rather than allowing the magnet to deform and then checking if strain is acceptable, components are designed from the outset to keep strain below the critical threshold of the superconductor. REBCO tapes can irreversibly degrade beyond about 0.4% tensile strain[39]. The Kronos *S.M.A.R.T.* tape architecture is an example where this philosophy is applied: the REBCO tape is integrated on a high-strength alloy substrate (MP35N, a cobalt-nickel alloy) that carries most of the mechanical load, and the coil is pre-compressed so that the superconducting layer never sees tension above 0.4% even at full field[44][45]. In practice, Kronos employs a carbon-fiber overwrap "cocoon" and a central bucking cylinder to put the coil under compressive preload[46]. When the magnets are energized, the Lorentz forces try to tension the coil, but they first have to overcome the built-in compression, thereby keeping the net strain in the brittle REBCO near zero. Such measures are crucial for any high-field spherical tokamak magnet, as the combination of high field and compact geometry would otherwise strain the superconductors beyond safe limits during operation or disruption events.

Thermal and fatigue considerations also enter the design. High-performance tapes and structures use materials that remain strong at cryogenic temperatures (for example, MP35N retains high yield strength at 4 K, unlike some steels which become brittle). The coil cases and supports must handle not only static forces but also dynamic loads from plasma disruptions. A disruption (a sudden loss of plasma current and pressure) induces huge eddy currents and *halo currents* in the structures, imparting impulsive forces. These transients can cause stresses even higher than steady-state loads. Therefore, spherical tokamak designers incorporate **fault scenarios** into the structural analysis: the support structure (like bucking cylinders, radial beams, etc.) is dimensioned to survive the worst-case asymmetric disruption. Often a factor of safety >~1.5 is used on yield stress for these cases. Additionally, **fatigue** is a concern if pulsed operation is planned. The NSTX Info Report highlighted that their centerstack casing had to endure thousands of pulses without cracking; similarly, a future ST reactor with superconducting coils would see many pulses or steady dwell under high stress, so materials like high-strength steels or composites with good fatigue endurance are favored.

One practical design compromise is between shaping and coil stress: pushing for more extreme shaping (higher κ, very negative δ) generally requires PF coils to be closer to the plasma and exert stronger forces, which in turn can stress the TF coils and structure more. For example, achieving κ = 3 instead of 2 might necessitate coils that almost hug the plasma surface, leaving less room for support structure and exposing them to higher neutron/heat flux. This reduces the engineering margin. Likewise, a highly negative δ plasma might require placing shaping coils on the outboard side in unusual positions, adding complexity to coil supports. Reactor studies (like a DEMO-shaped steady-state design) have noted that by dialing down triangularity slightly (less negative or less positive), one can ease the mechanical stresses with only a minor impact on plasma performance[19]. Thus, the "optimal" shape for a reactor is not necessarily the one with absolute highest ideal MHD stability, but one that balances stability gains against structural feasibility and coil lifetime.

In terms of **magnetic well depth**, although tokamaks don't have a "magnetic well" in the sense mirror machines do, one can think of the vertical stability as a potential well created by the conducting structures. A deeply shaped plasma with the right passive stabilizers can have a stronger restoring force (deeper well) for vertical displacements, but this usually means tighter clearance and more induction of image currents in the walls – again a trade-off since tighter walls mean less space for blankets or shield. Spherical tokamaks benefit from having the conducting vacuum vessel relatively close due to compactness, which actually helps vertical stability to some extent. Negative triangularity, when implemented in a double-null or near-double-null configuration, could potentially create a favourable distribution of current that aids vertical positioning (for instance, a quasi-double-null NT plasma could have some "magnetic well" vertically between the two X-points). These aspects are still being researched, but it's clear that structural design (where to put supports, how thick to make them) cannot be divorced from plasma shape decisions.

In summary, **mechanical design innovations are enabling the advanced shapes and high fields** needed for compact tokamaks. Using high-strength materials, pre-stressing methods, and supports like bucking cylinders, engineers can ensure coils survive the large forces (both steady and transient). The Kronos approach of a "strain-first" HTS magnet design[44] exemplifies the level of sophistication now being applied: rather than simply making coils as strong as possible, the

entire magnet system, from substrate to structure, is orchestrated to keep the superconductor in its safe mechanical envelope while delivering 20+ tesla fields. For future reactors, this means we can contemplate small, high-power plasma cores with extreme shaping that were previously thought unbuildable. The remaining piece is to ensure that this all works with the chosen fusion fuel cycle and power extraction scheme, which is the topic of the next section.

Relevance to Aneutronic Fuel Cycles

Next-generation fusion concepts are increasingly interested in aneutronic or semi-aneutronic fuel cycles—most notably D–^{3}He and p–^{11}B (proton–boron-11)—which produce much lower neutron flux than the conventional D–T cycle. These fuels pose higher requirements (higher temperature, confinement, and beta) but offer unique advantages if realized. The geometric and stability strategies discussed in this paper are particularly relevant to aneutronic reactors for several reasons:

1. Field-line geometry for charged-particle extraction: Aneutronic fusion reactions release their energy predominantly in **charged fusion products**. For D–^{3}He, the primary reaction yields a 3.6 MeV α particle and a 14.7 MeV proton; for p–^{11}B, three α particles (~2.9 MeV each) are produced. In a traditional tokamak, these charged particles would slow down in the plasma and transfer their energy to fuel ions and electrons (which is good for self-heating, but then the energy ends up as heat that must be extracted by a coolant). However, one can envision a reactor that **directly extracts the energy of these charged particles** before it thermalizes, using electric or magnetic methods. The shape and magnetic configuration of the plasma will dictate how and where these fusion alphas and protons escape. A spherical tokamak with carefully shaped flux surfaces could, for example, be designed so that a significant fraction of fast alphas are promptly lost to a specific region (e.g., out the divertor throat) where **direct energy conversion (DEC)** modules await. Negative triangularity may aid this by expanding the divertor and SOL regions, giving more space and a defined path for escaping particles. Moreover, with $\delta < 0$, the **magnetic mirror ratio** from core to divertor can be different, potentially allowing some high-energy ions to free-stream out once they reach the end of the machine. In essence, the shaping can create "funnels" in the magnetic field structure. As noted in one design study, clever field geometry can indeed channel a fraction of the charged exhaust into collector modules[47]. There, multi-stage electrostatic grids decelerate the ions, converting their kinetic energy directly into a DC electrical potential[48]. This concept, demonstrated in principle in the 1980s on mirror machines, can in theory achieve very high efficiency. The key is to have a quiescent plasma edge (no ELMs or disruptions) and a well-defined exhaust beam of fusion products – precisely what a stable NT spherical tokamak aims to provide.

2. Direct energy conversion systems: In addition to electrostatic collectors for ions, aneutronic reactors can employ **magnetohydrodynamic (MHD) generators** and other exotic schemes to harness energy. A compact, high-field tokamak is an excellent platform for this. For instance, the moving conductive fluid in a blanket (say, a liquid metal or molten salt used to absorb heat) can cut through the high magnetic field and act as an MHD generator, producing electric current as it flows[49]. The high fields (~20 T or more in some designs) make this effect stronger, potentially yielding significant direct power before the fluid even reaches a turbine. Similarly, as described in the Kronos design philosophy, one can "tune" the entire system to recover kinetic energy *before* it becomes random thermal energy[50]. In a negative triangularity ST, with no intense neutron bombardment, one can place functional surfaces very close to the plasma. This means **electrostatic plates, inductive pickup loops, or even thermionic converters** can be integrated right into the

first wall/divertor region to capture energy from charged particles and photons. For example, an array of biased rings in the divertor could act as a retarding field analyzer on steroids, slowing down escaping alphas/protons and collecting their charge as current. At the same time, the blanket flowing behind those plates sees a changing magnetic flux (from plasma transients and particle currents), generating inductive power. These direct conversion approaches have been explored for D–^{3}He tokamaks in past studies[51], and the consensus is that they could greatly boost net electrical efficiency – potentially 60% or higher conversion efficiency, roughly double that of a steam cycle[52][53]. The **magnetic geometry** must be designed to accommodate these; for instance, one might want slightly open field lines at the ends or special "cusp" regions. Tokamaks are typically closed field devices, but by using a divertor or limiter, we intentionally allow field lines to intersect material surfaces. That intersection is where DEC hardware would reside. Negative triangularity's large divertor could facilitate placing such hardware at an optimal angle or location where it intercepts a lot of fusion products without interfering with core plasma.

3. Reduced neutron load and compact shields: Aneutronic fuels drastically cut the neutron production (though not entirely zero for D–^{3}He or p–^{11}B due to side reactions). Even so, the neutron wall loading can be orders of magnitude lower than in a D–T reactor[54]. This allows a much thinner blanket/shield. In a D–T tokamak like ITER, nearly 1–2 m of shielding is required between the plasma and magnets to absorb 14 MeV neutrons and protect superconductors. In a D–^{3}He or p–^{11}B device, one might need only ~10 cm of shield (for secondary neutrons) or, conceivably, none at all if the neutron output is negligible. The benefit is that magnets can be placed very close to the plasma, enabling the ultra-compact designs envisaged in some private fusion roadmaps[55]. Spherical tokamaks already have a slender central column, and removing a thick blanket from that region means a higher field on axis for the same coil, or a smaller machine for the same fusion power. Thus, the combination of spherical aspect ratio and aneutronic fuels synergizes to shrink reactor size. Negative triangularity might further help by shaping the plasma so that any neutrons that are produced have a clear line-of-sight to small dedicated capture sections, rather than streaming into the magnets. Additionally, the lack of ELMs and disruptions in the envisioned operating regime means the first wall and direct conversion components can be designed for steady-state heat flux without having to survive giant transient spikes. This again is a more tractable engineering problem and dovetails with the aneutronic philosophy of "small, modular, and high-tech" rather than "brute-force heat handling."

4. Divertor and particle flow control: Even aneutronic plasmas produce "ash" – helium nuclei in the case of D–^{3}He or p–^{11}B. These must be exhausted to keep the plasma clean and to recover their energy. The divertor configuration in a high-shaped ST provides a way to constantly remove these alpha particles once they slow down (or even when still fast if using DEC). Magnetic shaping is critical here. For instance, one could employ a double-null configuration (top and bottom divertors) to efficiently channel particles regardless of drifts. Negative triangularity could be used on the bottom divertor to expand it for power handling, while perhaps a slight positive triangularity on top to help particle removal – essentially a mixed shape scenario if engineering allows. Furthermore, shaping influences the **SOL flow patterns** – the grad-B drift and curvature drift directions depend on triangularity sign. Some studies indicate that NT plasmas have different scrape-off layer width characteristics, possibly broader footprints, which could ease removal of helium ash. By tailoring the X-point position (e.g., moving it outboard or down), one can also control where the neutral gas pressure peaks in the divertor, aiding pumping. All these are knobs provided by geometry.

In summary, the advanced shaping strategies of high-κ, low-A, negative-δ are not only compatible with aneutronic operation, but actually beneficial. They enable a stable, high-pressure core needed for harder-to-burn fuels, and at the same time create a favorable environment to directly harvest charged fusion products and handle exhaust gently. The **lack of neutron burden** in these fuels means we can fully exploit the compact geometry: we don't need huge blankets and can integrate energy conversion tech close to the plasma. The wall becomes more than just a shield – it becomes an active part of the power cycle, as vividly described in the Kronos concept where "the outer wall…is partnering with [the plasma], co-generating power in almost the same breath that the plasma produces it"[52]. Negative triangularity's steady exhaust and spherical tokamak's high beta are enablers of this paradigm.

Risk Acknowledgment and Ongoing Research Disclosure

The plasma physics underpinning the S.M.A.R.T. reactor introduces several unprecedented challenges that the Kronos team recognizes and is actively addressing [1]. Operating an ultra-high-pressure, aneutronic plasma with advanced fuels (D–^{3}He, p–^{6}Li, p–^{11}B) at >100 million K in a compact, negative triangularity tokamak pushes well beyond the empirical database of conventional D–T devices [4]. We acknowledge the risk of plasma instabilities and uncertain confinement performance in this novel regime. To mitigate this, Kronos has implemented a multi-layered control and diagnostics program: an extensive array of graphene-based plasma sensors coats the first wall, providing real-time telemetry on fast ions, edge currents, and incipient instabilities [2]. These data feed into a proprietary AI control system that can adjust fueling, RF power, and coil currents on the fly to suppress instabilities and maintain optimized confinement. We are also leveraging full-scale 3D magnetohydrodynamic (MHD) simulations and machine-learning tools trained on historical confinement data to predict and avoid deleterious plasma modes before they grow. Fuel cycle integration risks – such as precise D–^{3}He mixture control, ash removal, and steady p–^{11}B ignition – are being tackled via a dedicated fuel research platform [3]. Small-scale plasma experiments (in collaboration with university labs) are underway to validate nuclear reaction rates and confinement behavior for our chosen fuels at relevant parameters, de-risking the transition from D–^{3}He ignition to p–B11 steady-state operation. The team is cognizant of the limited operational experience worldwide with D–^{3}He fusion; as a result, we have built redundant heating systems and robust divertor designs in case plasma performance falls short of predictions. Each heating and current-drive subsystem (NBI, dual-frequency RF, etc.) has a fail-safe overlap to ensure we can attain and sustain the required plasma conditions even if one method underperforms. We draw confidence from historical analogs: the mid-20th-century TMX and mirror experiments demonstrated that plasma stability could be maintained in non-standard configurations when guided by sufficient diagnostics and fast controls [7], and modern tokamaks like DIII-D have explored negative triangularity with promisingly improved stability [6]. These precedents inform our physics models but also temper our expectations – we have built significant margin into stability thresholds and engineered the device to tolerate excursions (e.g. by incorporating high-strength wall materials and flexible power exhaust paths [5]). Ongoing validation: Kronos's digital twin of the plasma – an integrated simulation incorporating transport, turbulence, and stability physics – is run continuously to benchmark our design against both historical data and upcoming experimental results from partner facilities. Any discrepancy between predicted and observed plasma behavior will trigger a design review and model update, ensuring our confinement predictions remain grounded. In parallel, a proprietary control system testbed (using a smaller-scale plasma column and virtual reality feedback) is de-risking the

actuator response times required for S.M.A.R.T.'s active control. Looking ahead to 2030, we commit to publishing a full report on plasma performance metrics achieved in prototype tests, including energy confinement time, beta limits, and stability margins compared to design values. Should any plasma physics risk remain open at that time, its mitigation plan and remaining uncertainties will be transparently detailed. Kronos's approach is one of humility and scientific rigor: we acknowledge that our confinement and fuel cycle strategy must earn validation through data. All early results, whether confirming or challenging our models, will feed back into the design. By maintaining this continuous R&D loop through the rest of the decade, we aim to retire the major plasma physics risks well before the 2032 commercial rollout, with peer-reviewed publications documenting the achieved confinement, stability, and burn control in our system.

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Conclusion

Advanced plasma shaping is proving to be a cornerstone for achieving stable, high-performance operation in next-generation tokamaks. In this review, we examined how compact spherical tokamaks, enhanced by high elongation and negative triangularity geometry, can confine plasmas at high normalized beta while mitigating some of the most problematic instabilities. We highlighted that by inverting the traditional D-shape (negative δ), experiments have obtained H-mode level confinement without ELMs[1], an achievement with enormous implications for reactor design. The spherical aspect ratio (A ~ 1.5–2) further boosts stability and confinement, allowing record beta values (β_T ~ 40% in NSTX) when combined with strong shaping and active control[56]. The key trade-offs – vertical stability, coil stresses, and limited space – can be managed with modern engineering solutions like bucking cylinder supports and robust HTS magnet constructions. Maintaining the plasma shape under load has driven innovations such as the strain-first magnet design, as seen in the Kronos tape architecture, ensuring HTS coils remain within safe strain (<0.4%) even under extreme fields[44].

Negative triangularity in particular emerges as a "game changer" for future reactors, as one experimenter put it[5]. By operating in this shape, a reactor could avoid pulsed heat loads (ELMs) and distribute exhaust power over a larger area, simplifying the task of power handling. At the same time, stable high-beta operation can be sustained, since NT does not fundamentally lower the beta limit within the parameter space explored – it simply changes how the plasma reaches that limit. The combination of **high β and steady-state operation** is especially critical for aneutronic fusion, where every bit of efficiency counts. In aneutronic concepts, the fusion energy can be extracted through direct means (charged particles and radiation) rather than solely thermal

routes[9]. We discussed how a compact NT spherical tokamak is well-suited to this: it confines a hot plasma with minimal dilution (thanks to efficient ash removal via a well-shaped divertor) and allows placing energy conversion structures close to the action (thanks to low neutron output and stable plasma-wall interfaces). The result could be a fusion reactor with unprecedented efficiency and power density – a machine that turns fusion products directly into electricity with minimal intermediate steps[47][57].

In conclusion, geometry and stability considerations will heavily influence the viability of next-generation fusion reactors. Strategies such as strong plasma shaping ($\kappa > 2$, negative δ), low aspect ratio design, and high-field HTS magnets must work in unison. The evidence so far suggests they can: experiments on devices like DIII-D, TCV, and NSTX-U, and designs like SMART and Kronos, indicate that we can have plasmas that are **highly pressurized yet quiescent**, with the shape helping to tame instabilities that once seemed inevitable[2]. The challenge now is to integrate these concepts into a coherent reactor system. The path forward will involve continued refinement of shaping control (possibly using AI-driven feedback as shapes become complex), further materials development for magnets and plasma-facing components, and full-power testing of these ideas in upcoming devices. The promise, however, is compelling – a compact, efficient fusion reactor that leverages geometry to unlock performance, operating stably on advanced fuels and delivering power in new, efficient ways. Geometry, long an afterthought in some older designs, is moving to center stage as a tool to **engineer out** the obstacles on the road to fusion energy. The successes in high-beta, negative triangularity confinement are an encouraging sign that this approach is the right one for the reactors of the near future.

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CHARTS

Charts 1: Plasma shaping and beta performance. (a) Comparison of normalized beta (β) versus normalized current (I_n/NA) for different plasma cross-sections—circular, D-shape, and racetrack—showing the higher stability limits achievable with optimized shaping. (b) Experimental database of beta versus I_paB across major tokamaks (JET, ITER, DIII-D, PBX-M, ASDEX, etc.), illustrating the path toward advanced high-β operation and confinement scaling. 7
Charts 2: Magnetic flux surface contours for compact, high-elongation plasmas with negative triangularity. The color maps illustrate normalized poloidal field strength (Bθ) across various shaping configurations, showing how negative triangularity modifies edge stability, reduces ELM activity, and supports enhanced confinement in the S.M.A.R.T. geometry. 9

FIGURES

Figure 1: Engineering solutions in compact low-A magnets: bucking cylinder, coil wedge supports, and pre-tensioning elements that accommodate high Lorentz forces and minimize strain at 30–35 T operation. 5
Figure 2: Negative triangularity plasma configuration in a compact spherical tokamak. The inverted-D cross-section improves edge stability, suppresses ELMs, and maintains high-β confinement suitable for aneutronic fuel operation. 10
Figure 3: Divertor heat-flux footprint for negative triangularity versus positive triangularity plasmas. The strike-point profile in $\delta < 0$ cases is broader and the peak heat load reduced, improving divertor survivability at reactor-relevant power exhaust conditions. 10

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Geometry and Stability in the Compact Spherical Tokamak

Negative Triangularity and High-Beta Confinement

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Detailed Technical Appendix Available Upon Request
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